

Amirali Abedi

europass

Phone number: (+40) 773859844 (Mobile) **Email address:** AmiraliAbedi97@gmail.com

Website: https://github.com/Amir-Abedi **LinkedIn:** Amirali Abedi

ABOUT ME

Second-year MSc student in Telecommunication Engineering at Politehnica University Timișoara, focused on deep learning for multimedia processing (images, video and audio). First author of two papers accepted at ISETC 2026 (with Prof. Mugurаș Mocofan) on lightweight CNN-based super-resolution and image restoration for resource-constrained devices. Hands-on experience with PyTorch, TensorFlow, Transformers, diffusion models and deployment on NXP microcontrollers. Seeking a research internship in 2027 as a stepping stone toward a PhD in deep learning for vision and audio.

EDUCATION AND TRAINING

29/09/2025 – CURRENT Timisoara, Romania

MASTER OF SCIENCE IN ELECTRONICS, TELECOMMUNICATIONS AND INFORMATION TECHNOLOGY (COMMUNICATION NETWORKS) Politehnica University Timișoara

Expected graduation: June 2027. Average grade: 9.09/10.
- Deep learning for image processing: CNNs
- Transformer-based models for speech, such as WavLM
- Generative models: GANs and diffusion models
- Research projects, one per semester: image classification with ResNet, DenseNet and MobileNet (semester 1); CNN-based super-resolution (semester 2); object detection with YOLO (semester 3, ongoing)
- Coursework in Pattern Recognition, with an introduction to NLP
- Deployment of lightweight neural networks on microcontrollers (NXP, Arduino, Embedded C++)

Level in EQF EQF level 7

08/05/2026 – 07/06/2026 Valmiera, Latvia

ERASMUS+ BLENDED INTENSIVE PROGRAMME: XR MEETS HEALTH TECH Vidzemes Augstskola (ViA)

Erasmus+ Blended Intensive Programme (3 ECTS): "XR meets Health Tech: Investigating Virtual and Augmented Realities in the Age of AI".
- Medical applications of AI and digital health solutions
- Teamwork in an international group of students

PUBLICATIONS

2026

Patch-Based Super-Resolution Using Lightweight Convolutional Neural Networks on Resource-Constrained Microcontrollers

First author. Accepted at ISETC 2026; not yet published.

Abedi, A., Mocofan, M. (2026). Patch-Based Super-Resolution Using Lightweight Convolutional Neural Networks on Resource-Constrained Microcontrollers. ISETC 2026. Accepted, not yet published.

Authors: Amirali Abedi, Mugurаș Mocofan | **Journal Name**: 2026 International Symposium on Electronics and Telecommunications (ISETC) | **Publisher**: IEEE (forthcoming)

2026

Adaptive Image Restoration via Enhanced Multi-Layer SRCNN Architectures

First author. Accepted at ISETC 2026; not yet published.

Abedi, A., Mocofan, M. (2026). Adaptive Image Restoration via Enhanced Multi-Layer SRCNN Architectures. ISETC 2026. Accepted, not yet published.

Authors: Amirali Abedi, Mugurаș Mocofan | **Journal Name**: 2026 International Symposium on Electronics and Telecommunications (ISETC) | **Publisher**: IEEE (forthcoming)

SKILLS

Programming & Tools

Python (computer programming) | C++ | MATLAB | Java (computer programming) | Git/Github, Docker, Gitlab | Unix-based OS | Jupyter Notebook, Google Colab, Kaggle Notebook, VS Code and PyCharm | LaTeX

Computer Vision & Machine Learning

Computer Vision | Deep Learning | Artificial Intelligence | Machine Learning | Digital Image Processing | TensorFlow | PyTorch | Keras | Image super resolution | Image analysis | Image Segmentation | ResNet, DenseNet, MobileNet | Deep Neural Networks (CNNs, GANs) | Object Detection (SSD & YOLO) | CNN and RNN | GAN (Generative Adversarial Network) | Anomaly Detection (VAE, AE) | Scikit-Learn | Numpy | Diffusion Models | Pandas | Hugging Face Transformers

Embedded Systems

embedded systems | Microcontroller Programming (Arduino, PIC, NXP) | Embedded C++

LANGUAGE SKILLS

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C1
FRENCH	A2	B1	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CERTIFICATIONS

Romanian Space Initiative (ROSPIN), 28/05/2026

ROSPIN Academy – Level 3

Online course (7 lectures and quizzes) covering introductory space concepts, systems engineering and detailed subsystem design.

Mode of learning: Online

IEEE Education Society / Politehnica University Timișoara, 26/06/2026

DEMOcon Summer School 2026 – Digital Education Ecosystems: MOOCs, AI and Microcredentials

Certificate of completion (30 hours of activity, 1 ECTS) for the IEEE DEMOcon Summer School, held at Politehnica University Timișoara, 21–24 June 2026. Topics: experiences and developments in digital education, MOOCs, AI in education and microcredentials.

Mode of learning: Presential

Link https://democon-conference.org/2026